



Complete Genome Sequences of *Mycobacterium* Phages Tripl3t and Zeuska

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ABSTRACT Tripl3t and Zeuska are siphoviral bacteriophages that were isolated from *Mycobacterium smegmatis* mc² 155 and contain double-stranded DNA genomes 53,565 bp and 53,598 bp in length, respectively. Tripl3t and Zeuska were annotated by students at Bluff Dale High School (Bluff Dale, TX) and Tolar High School (Tolar, TX) in community engagement with Tarleton State University.

Mycobacteriophages from the Howards Hughes Medical Institute Science Education Alliance-Phage Hunters Advancing Genomics and Evolutionary Science (SEA-PHAGES) library have been used previously to successfully treat *Mycobacterium* infections (1, 2). As part of the SEA-PHAGES program, we report the genomes of *Siphoviridae* mycobacteriophages Tripl3t and Zeuska, which were isolated from soil samples from Washington, DC (38.92426N, 77.01955W), and Providence, Rhode Island (41.824047N, 71.403114W), respectively. Mycobacteriophage Tripl3t was originally isolated by Demi I. F. Lewis at Howard University (Washington, DC), and mycobacteriophage Zeuska was originally isolated by Emma Herold at Brown University (Providence, RI). The samples were incubated in 7H9 liquid medium at 37°C for 2 h before the supernatant was centrifuged, filtered through a 0.22-μm filter, and incubated with *Mycobacterium smegmatis* mc² 155 at 37°C on Luria agar plates. Bacteriophages were isolated by two rounds of picking a single, well-separated plaque, followed by dilution of samples in a 10-fold dilution series and plating with *M. smegmatis* mc² 155 (3). Tripl3t formed large bullseye plaques, while Zeuska formed medium plaques with cloudy borders (Fig. 1). High-titer lysates were obtained by flooding “webbed” plates, as described in the Phage Discovery Guide, and DNA was extracted using the Promega Wizard DNA clean-up system (3). High-titer lysates (with dimethyl sulfoxide at a final concentration of 6.7%) were shipped on dry ice to and from the Pittsburgh Bacteriophage Institute and stored at –80°C. The Tripl3t genomic library was prepared by Virginia Commonwealth University Nucleic Acids Research Facilities using the 454 DNA library preparation kit and was sequenced with a Roche 454 GS FLX sequencer to approximately 795-fold coverage from 104,617 total reads (average read length, 165 bp). The Zeuska genomic library was prepared using the TruSeq DNA nanokit and was sequenced with an Illumina MiSeq system at the Pittsburgh Bacteriophage Institute to approximately 3,844-fold coverage from 749,005 total reads (average read length, 150 bp) (4). Raw reads were assembled to produce single-bacteriophage contigs using Newbler v2.7, and Consed v22.0 was used to check for completeness, accuracy, and genome termini (4, 5). Tripl3t and Zeuska have linear double-stranded genomes of 53,565 bp and 53,598 bp, respectively. Both bacteriophage genomes have a G+C content of 63.7% and a 10-base 3′ sticky overhang with the sequence 5′-CGGATGGTAA-3′.

Whole-genome nucleotide alignments by BLASTn (<https://blast.ncbi.nlm.nih.gov>) showed Tripl3t and Zeuska to have close nucleotide identity to subcluster A1 bacteriophages Wheeler (GenBank accession number [NC_022070](https://www.ncbi.nlm.nih.gov/nuclot/NC_022070)) and Abrogate (GenBank

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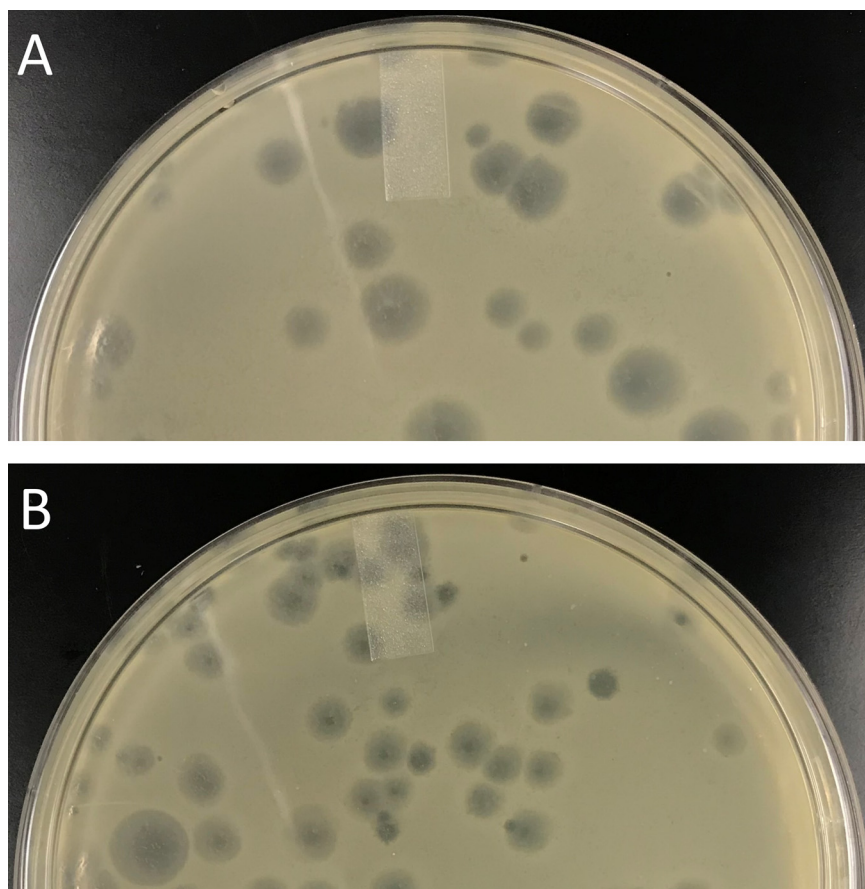


FIG 1 Images of plaques from bacteriophages Tripl3t (A) and Zeuska (B) incubated with *Mycobacterium smegmatis* mc² 155 on Luria agar plates. After 24 h of incubation at 37°C, Tripl3t formed mostly large bullseye plaques and Zeuska formed medium plaques with cloudy borders.

accession number [KM597531](#)) (6–8). Initial autoannotations were generated using Glimmer v3.02 (9) and GeneMark v2.5p (10), with manual revisions using DNA Master v5.23.2 (<http://phagesdb.org/DNAMaster>) and PECAAN (<https://pecaan.kbrinsgd.org>). No tRNA genes were detected by ARAGORN v1.2.38 (11) or tRNAscan-SE v2.0 (12). Putative gene functions were assigned with HHpred v3.0beta (13, 14) and BLASTp (8). All tools were run with default parameters. Putative functions were assigned to 36 of 91 predicted protein-coding genes for Tripl3t and 35 of 95 predicted protein-coding genes for Zeuska. Both bacteriophage genomes contain genes for virion assembly and structure, lysis proteins, host integration and excision proteins, DNA primase, RusA-like resolvase, RtcB-like ligase, integrase, and an immunity repressor.

Data availability. GenBank and SRA accession numbers are as follows: Tripl3t, GenBank accession number [MK524499](#) and SRA accession number [SRX4721441](#); Zeuska, GenBank accession number [MK524506](#) and SRA accession number [SRX4721438](#).

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